

EXP:15

Determination and plotting of Inductance (i) Self-inductance and Mutual inductance of current carrying coil, (ii) solenoid and toroid with different (a) length and (b) circumferential area.

INPUT:

```
Enter the turns of self inductance: 100
Enter the phi value: 3*10^-2
Enter the current flowing in inductance: 3
"Self inductance:"
1.
Enter the primary turn: 100
Enter the secondary turn: 300
Enter the phi 12: 3*10^-4
Enter the phi 21: 4*10^-6
Enter the CURRENT 1: 3
Enter the CURRENT 2: 2
"Mutual inductance M12:"
0.0300000
"Mutual inductance M21:"
0.0002000
Enter the length value is: 0.003
Enter the radius value is: 0.004
"Solenoid Inductance:"
0.001256
"Toroid Inductance:"
0.00015
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```

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exp333.sci (C:\Users\pavan\Documents\beee\exp333.sci) - SciNotes
*exp333.sci X
1 clc;
2 clear;
3 n = input("Enter the turns of self inductance:");
4 phi = input("Enter the phi value:");
5 i = input("Enter the current flowing in inductance:");
6 L = n * phi / i;
7 disp("Self inductance:", L);
8 n1 = input("Enter the primary turn:");
9 n2 = input("Enter the secondary turn:");
10 O12 = input("Enter the phi 12:");
11 O21 = input("Enter the phi 21:");
12 i1 = input("Enter the CURRENT 1:");
13 i2 = input("Enter the CURRENT 2:");
14 M12 = n2 * O12 / i1;
15 M21 = n1 * O21 / i2;
16 disp("Mutual inductance M12:", M12);
17 disp("Mutual inductance M21:", M21);
18 LENGTH = input("Enter the length value is:");
19 Radius = input("Enter the radius value is:");
20 solenoid = (4 * 3.14 * 10^-7 * i) / LENGTH;
21 toroid = (4 * 3.14 * 10^-7 * i) / (2 * 3.14 * Radius);
22 disp("Solenoid Inductance:", solenoid);
23 disp("Toroid Inductance:", toroid);
24 a = linspace(0, 10, 50);
25 b = L * exp(-0.1 * a);
26 c = M12 * exp(-0.1 * a);
27 d = M21 * exp(-0.1 * a);
28 e = solenoid * exp(0.1 * a);
29 f = toroid * exp(0.1 * a);
30 figure();
31 subplot(2, 2, 1);
32 plot2d3(a, b);
33 xlabel("TIME ./ VARIABLE (a)");
34 ylabel("SELF INDUCTANCE");
35 title("SELF INDUCTANCE");
36 subplot(2, 2, 2);
```

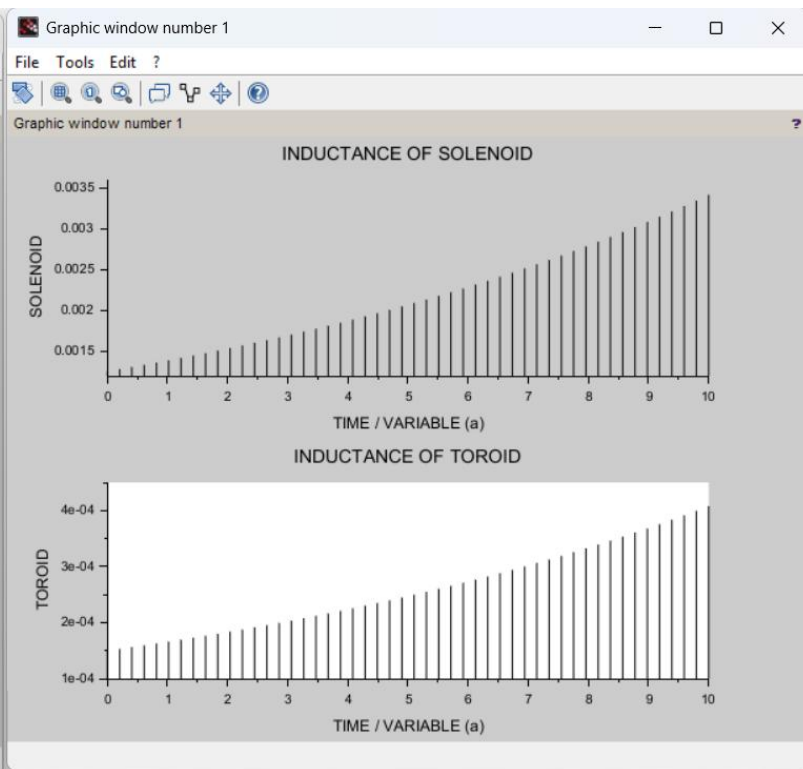
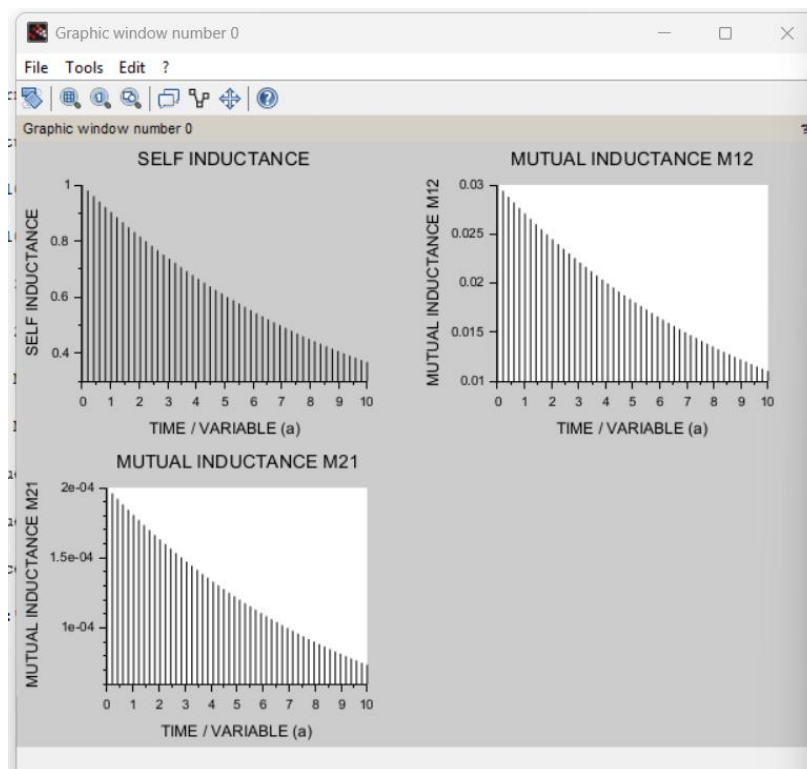
Line 56, Column 30.

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37 plot2d3(a, c);-
38 xlabel("TIME ./ VARIABLE (a)");-
39 ylabel("MUTUAL-INDUCTANCE-M12");-
40 title("MUTUAL-INDUCTANCE-M12");-
41 subplot(2, 2, 3);-
42 plot2d3(a, d);-
43 xlabel("TIME ./ VARIABLE (a)");-
44 ylabel("MUTUAL-INDUCTANCE-M21");-
45 title("MUTUAL-INDUCTANCE-M21");-
46 figure();-
47 subplot(2, 1, 1);-
48 plot2d3(a, e);-
49 xlabel("TIME ./ VARIABLE (a)");-
50 ylabel("SOLENOID");-
51 title("INDUCTANCE-OF-SOLENOID");-
52 subplot(2, 1, 2);-
53 plot2d3(a, f);-
54 xlabel("TIME ./ VARIABLE (a)");-
55 ylabel("TOROID");-
56 title("INDUCTANCE-OF-TOROID");|

```

Graphs:



Calculation:

Exp: 15

Self inductance = $N = 100$

Phi value = 3×10^{-2}

Current flowing in inductance = 3 A

Primary turn = 100

Secondary turn = 300

$\phi_{12} = 3 \times 10^{-4}$

$\phi_{21} = 4 \times 10^{-6}$

Current 1 = 3

Current 2 = 2

Length $l = 3 \text{ mm} = 0.003$

radius = $4 \text{ mm} = 0.004$

A) i) self Inductance (L)

Inputs:-

Turns (N) = 100

magnetic flux (ϕ) = 3×10^{-2}

Current (i) = 3

Formula:- $L = \frac{N \phi}{i}$

$$L = \frac{100 \times (3 \times 10^{-2})}{3} = \frac{3}{3} = 1 \text{ H} //$$

(ii) Mutual inductance (M_{12} and M_{21})

Inputs:-

$N_1 = 100$, $N_2 = 300$

$\phi_{12} = 3 \times 10^{-4}$, $\phi_{21} = 4 \times 10^{-6}$

$i_1 = 3$, $i_2 = 2$

Formula for M_{12} :-

$$M_{12} = \frac{n_2 \cdot \phi_{12}}{i_1}$$

$$= \frac{300 \times (3 \times 10^{-4})}{3} = \frac{0.09}{3} = 0.003 \text{ H.}$$

Formula for M_{21} :-

$$M_{21} = \frac{n_1 \cdot \phi_{21}}{i_2}$$

$$= \frac{100 \times (4 \times 10^{-6})}{2} = \frac{4 \times 10^{-4}}{2} = 2 \times 10^{-4} = 0.0002 \text{ H.}$$

iii) Solenoid and Toroid Inductance

Input:-

- current (i) = 3
- Length (L) = 0.003
- Radius (R) = 0.004
- Permeability of free space (μ_0) = $4\pi \times 10^{-7} \text{ Tm/A}$ $\approx 1.2566 \times 10^{-6}$

Formula for Solenoid:-

$$L = \frac{\mu_0 \cdot N^2 \cdot A}{l}$$

$$\frac{i \times \mu_0}{L} \text{ used in scalar calc.}$$

$$\text{Solenoid} = \frac{4\pi \times 10^{-7} \times i}{L} \Rightarrow \frac{4 \times 3.14 \times 10^{-7} \times 3}{0.003} \Rightarrow \frac{3.768 \times 10^{-6}}{0.003}$$

$$= 0.001256 \text{ H.}$$

$$\text{Toroid} = \frac{4\pi \times 10^{-7} \cdot i}{2\pi \times R} = \frac{4 \times 3.14 \times 10^{-7} \times 3}{2 \times 3.14 \times 0.004}$$

$$= \frac{3.768 \times 10^{-6}}{0.02512} = 0.00015 \text{ H.}$$